Solutions to Select Problems From *Conway. Functions of One Complex Variable. 2nd ed.*

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November 2, 2026

Preface

This document is a compilation of exercises I have completed from the second edition of John B. Conway's *Functions of One Complex Variable*. There are several motivations behind this project. First, I was unable to take the complex analysis course offered at my undergraduate institution. Second, complex analysis is important in describing the underlying mathematics behind quantum field theory, which I am interested in pursuing in graduate studies. Third, I am interested in learning about the mathematical theory of complex analysis, in addition and in contrast to the theory of real analysis.

The textbook was chosen with an emphasis on rigor, as I have found that having a strong mathematical foundation for a topic is invaluable when utilizing the theory in physics applications. I went through several other textbooks before settling on Conway's book; I first learned of this book through the excellent University of Chicago Undergraduate Mathematics Bibliography.

I found that the topics of most interest in their physical application tended to the middle and latter portion of the book. Therefore, the most effort was put into these chapters. The exercises shown here were not chosen in any systematic fashion.

As of this writing, the project remains in progress.

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Chapter II: Metric Spaces and the Topology of $\mathbb{C}$

§1: Definitions and Examples of Metric Spaces

1. Show that each of the examples of metric spaces given below is, indeed, a metric space. The four conditions for a metric space (X, d) and $x, y, z \in X$ are listed below:

- $d(x, y) \geq 0$
- $d(x, y) = 0$ iff $x = y$
- $d(x, y) = d(y, x)$
- $d(x, z) \leq d(x, y) + d(y, z)$

Also, describe $B(x; r)$ for each of these examples.

(a) For a metric space (X, d) with $Y \subseteq X$, consider the metric space (Y, d).

Proof. Take $x, y, z \in Y$. Then by definition $x, y, z \in X$, and by (X, d) as a metric space, must satisfy the four necessary conditions for (Y, d) to constitute a metric space since d is unchanged. The open ball $B(x; r)$ can be found as $B(x; r) = B_X(x; r) \cap Y$ where $B_X(x; r)$ is the open ball of X at x. ■

(b) Let $X = \mathbb{C}$ and define $d(x + iy, a + ib) = |x - a| + |y - b|$. Then consider the metric space $(\mathbb{C}, d)$.

Proof. The first condition is satisfied by definition of the absolute value. We find again by the definition of the absolute value that

$$\begin{aligned} d(x + iy, a + ib) = 0 &\iff |x - a| + |y - b| = 0 \\ &\iff |x - a| = 0 \wedge |y - b| = 0 \\ &\iff x + iy = a + ib \end{aligned}$$

and hence satisfy the second condition. The third condition follows immediately by the symmetry of the absolute value. To show the triangle inequality, take $x + iy, a + ib, c + id \in \mathbb{C}$. Then

$$\begin{aligned} &d(x + iy, a + ib) + d(a + ib, c + id) - d(x + iy, c + id) \\ &\quad = |x - a| + |y - b| + |a - c| + |b - d| - |x - c| - |y - d| \\ &\quad = (|x - a| + |a - c| - |x - c|) + (|y - b| + |b - d| - |y - d|) \\ &\quad \geq 0 + 0 = 0 \end{aligned}$$

by the triangle inequality for real numbers. The open ball $B(x;r)$ can be understood as a square of side length $\sqrt{2}r$ centered at x and rotated by $\pi/4$ radians in the complex plane, with the boundary not included. ■

(c) Let $X = \mathbb{C}$ and define $d(x+iy, a+ib) = \max\{|x-a|, |y-b|\}$. Then consider the metric space $(\mathbb{C}, d)$.

Proof. The first condition is satisfied by absolute value as non-negative, and the second condition follows readily by observing that $\max\{|x-a|, |y-b|\} = 0$ is achieved only with both arguments in the max function as 0, implying that $x+iy = a+ib$. Symmetry follows from the symmetry of the absolute value. To show the last condition, suppose without loss of generality that $|x-c| \geq |y-d|$. Then

$$\begin{aligned} &d(x+iy, a+ib) + d(a+ib, c+id) - d(x+iy, c+id) \\ &\quad = \max\{|x-a|, |y-b|\} + \max\{|a-c|, |b-d|\} - \max\{|x-c|, |y-d|\} \\ &\quad \geq \max\{|x-a|, |y-b|\} + \max\{|a-c|, |b-d|\} - |x-c| \\ &\quad \geq |x-a| + |a-c| - |x-c| \geq 0 \end{aligned}$$

This argument clearly functions analogously if $|x-c| \leq |y-d|$ and hence we have shown the triangle inequality follows in all possible cases. The open ball $B(x;r)$ is a square of side length $2r$ centered at x with boundaries excluded. ■

(d) Let X be any set and define $d(x,y) = 0$ if $x = y$ and $d(x,y) = 1$ if $x \neq y$. Consider metric space (X, d).

Proof. The first two conditions are satisfied immediately by the definition of d. Symmetry follows readily from inspection. The triangle inequality and $B(x;r)$ are described in the text. ■

(e) Let $X = \mathbb{R}^n$ and for $x = (x_1, \ldots, x_n), y = (y_1, \ldots, y_n)$ in $\mathbb{R}^n$ define

$$d(x,y) = \left[\sum_{j=1}^{n} (x_j - y_j)^2\right]^{\frac{1}{2}}$$

Consider the metric space $(\mathbb{R}^n, d)$.

Proof. The summation is of non-negative terms which must produce a corresponding non-negative square root thereby showing the first condition. The summation and hence square root is strictly positive unless $x_j = y_j$ for all $j \in \{1, \ldots, n\}$, which is equivalent to $x = y$, showing the second condition. Symmetry follows from the square in the summation, since the value $(x_j - y_j)$ is transformed to its additive inverse upon exchange of x and y. To show the triangle inequality, take $x, y, z \in \mathbb{R}^n$. We will proceed by induction. For the base case, we have $n = 1$, such that $x = x_1$, $y = y_1$, and $z = z_1$. We have

$$\begin{aligned} & d(x,y) + d(y,z) - d(x,z) \\ & \quad = \left[(x_1 - y_1)^2\right]^{\frac{1}{2}} + \left[(y_1 - z_1)^2\right]^{\frac{1}{2}} - \left[(x_1 - z_1)^2\right]^{\frac{1}{2}} \\ & \quad = |x - y| + |y - z| - |x - z| \geq 0 \end{aligned}$$

by the triangle identity for real numbers, so that the base case is satisfied. Next, assume for some $k \in \mathbb{Z}^+$ that $d(x,y) + d(y,z) - d(x,z) > 0$ where $x, y, z \in \mathbb{R}^k$, i.e.

$$\left[\sum_{j=1}^{k}(x_j - y_j)^2\right]^{\frac{1}{2}} + \left[\sum_{j=1}^{k}(y_j - z_j)^2\right]^{\frac{1}{2}} - \left[\sum_{j=1}^{k}(x_j - z_j)^2\right]^{\frac{1}{2}} \geq 0$$

For any $a, b, c \in \mathbb{R}$, where $a, b, c, \geq 0$, we have

$$\sqrt{a} + \sqrt{b} \geq c \iff a + b + 2\sqrt{ab} \geq c \tag{1}$$

since the square and square root are monotonically increasing functions on $\mathbb{R}^{\geq 0}$. It therefore follows that

$$
\begin{aligned}
0 &\leq \sum_{j=1}^{k}(x_j - y_j)^2 + \sum_{j=1}^{k}(y_j - z_j)^2 \\
&\quad + 2\sqrt{\sum_{j=1}^{k}(x_j - y_j)^2}\sqrt{\sum_{j=1}^{k}(y_j - z_j)^2} - \sum_{j=1}^{k}(x_j - z_j)^2 \\
&\leq \sum_{j=1}^{k+1}(x_j - y_j)^2 + \sum_{j=1}^{k+1}(y_j - z_j)^2 + 2|x_{k+1} - y_{k+1}||y_{k+1} - z_{k+1}| \qquad (2) \\
&\quad + 2\sqrt{\sum_{j=1}^{k}(x_j - y_j)^2}\sqrt{\sum_{j=1}^{k}(y_j - z_j)^2} - \sum_{j=1}^{k+1}(x_j - z_j)^2 \\
&\leq \sum_{j=1}^{k+1}(x_j - y_j)^2 + \sum_{j=1}^{k+1}(y_j - z_j)^2 \qquad (3) \\
&\quad + 2\sqrt{\sum_{j=1}^{k+1}(x_j - y_j)^2}\sqrt{\sum_{j=1}^{k+1}(y_j - z_j)^2} - \sum_{j=1}^{k+1}(x_j - z_j)^2 \\
&\iff \left[\sum_{j=1}^{k+1}(x_j - y_j)^2\right]^{\frac{1}{2}} + \left[\sum_{j=1}^{k+1}(y_j - z_j)^2\right]^{\frac{1}{2}} - \left[\sum_{j=1}^{k+1}(x_j - z_j)^2\right]^{\frac{1}{2}} \geq 0 \\
&\iff d(x,z) \leq d(x,y) + d(y,z)
\end{aligned}
$$

where Equation 2 follows from a sum of the form of the right-hand side of Equation 1 with $a = (x_{k+1} - y_{k+1})^2$, $b = (y_{k+1} - z_{k+1})^2$, and $c = (x_{k+1} - z_{k+1})^2$, and Equation 3 follows from the deduction

$$0 \leq \left(\sqrt{\sum_{j=1}^{k}(x_j - y_j)^2}|y_{k+1} - z_{k+1}| - \sqrt{\sum_{j=1}^{k}(y_j - z_j)^2}|x_{k+1} - y_{k+1}| \right)^2$$

$$\iff \sum_{j=1}^{k}(x_j - y_j)^2 \sum_{j=1}^{k}(y_j - z_j)^2 +$$

$$2\sqrt{\sum_{j=1}^{k}(x_j - y_j)^2}\sqrt{\sum_{j=1}^{k}(y_j - z_j)^2}|x_{k+1} - y_{k+1}||y_{k+1} - z_{k+1}| +$$

$$(x_{k+1} - y_{k+1})^2(y_{k+1} - z_{k+1})^2 \leq$$

$$\sum_{j=1}^{k}(x_j - y_j)^2 \sum_{j=1}^{k}(y_j - z_j)^2 + |y_{k+1} - z_{k+1}|^2 \sum_{j=1}^{k}(x_j - y_j)^2 +$$

$$|x_{k+1} - y_{k+1}|^2 \sum_{j=1}^{k}(y_j - z_j)^2 + (x_{k+1} - y_{k+1})^2(y_{k+1} - z_{k+1})^2$$

$$\iff 0 \leq \sqrt{\sum_{j=1}^{k}(x_j - y_j)^2 + (x_{k+1} - y_{k+1})^2}\sqrt{\sum_{j=1}^{k}(y_j - z_j)^2 + (y_{k+1} - z_{k+1})^2}$$

$$- \sqrt{\sum_{j=1}^{k}(x_j - y_j)^2}\sqrt{\sum_{j=1}^{k}(y_j - z_j)^2} - |x_{k+1} - y_{k+1}||y_{k+1} - z_{k+1}|$$

$$\iff 0 \leq \sqrt{\sum_{j=1}^{k+1}(x_j - y_j)^2}\sqrt{\sum_{j=1}^{k+1}(y_j - z_j)^2} - \sqrt{\sum_{j=1}^{k}(x_j - y_j)^2}\sqrt{\sum_{j=1}^{k}(y_j - z_j)^2}$$

$$- |x_{k+1} - y_{k+1}||y_{k+1} - z_{k+1}|$$

By induction, the triangle inequality then holds for arbitrary $n \in \mathbb{Z}^+$. The open ball $B(x; r)$ is an n-sphere of radius r. ■

Chapter III: Elementary Properties and Examples of Analytic Functions

§2: Analytic Functions

1. Show that $f(z) = |z|^2 = x^2 + y^2$ has a derivative only at the origin.

Proof. First we fix $z = 0$. Taking $h = a + bi \in \mathbb{C}$,

$$f'(0) = \lim_{h\to 0} \frac{f(h) - f(0)}{h} = \lim_{h\to 0} \frac{a^2 + b^2}{a + bi} = \lim_{h\to 0} a - bi = \lim_{h\to 0} \bar{h} = 0$$

where we have used $h \to 0 \iff \bar{h} \to 0$ by inspection of the real and imaginary components. Now take $z = c + id \in \mathbb{C} \setminus \{0\}$. We will attempt to find the limit corresponding to the derivative by choosing two distinct paths – along the real axis ($h = a$), and along the imaginary axis ($h = ib$); the values should be identical if the derivative exists. Starting along the real axis,

$$\lim_{a\to 0} \frac{f(z + a) - f(z)}{a} = \lim_{a\to 0} \frac{f(a + c + id) - f(c + id)}{a} \tag{4}$$

$$= \lim_{a\to 0} \frac{a^2 + 2ac + c^2 + d^2 - c^2 - d^2}{a} \tag{5}$$

$$= \lim_{a\to 0} a + 2c = 2c \tag{6}$$

Analogously,

$$\lim_{b\to 0} \frac{f(z + bi) - f(z)}{bi} = \lim_{b\to 0} \frac{f(c + (b + d)i) - f(c + id)}{bi} \tag{7}$$

$$= \lim_{b\to 0} \frac{c^2 + b^2 + 2bd + d^2 - c^2 - d^2}{bi} \tag{8}$$

$$= \lim_{b\to 0} -(b + 2d)i = -2di \tag{9}$$

We have equality if and only if $c = d = 0$, so that $z = 0$, which is the case we already treated. Hence, f is differentiable only at the origin. ∎

2. Prove that if a_n, b_n are real and positive and $0 < b = \lim b_n, a = \limsup a_n$, then $ab = \limsup(a_n b_n)$. Does this remain true if the requirement of positivity is dropped?

Proof. Fix $\epsilon > 0$. Without loss of generality take $\epsilon < \min(a, b)$. Moreover, take $\epsilon' = \min(\frac{\sqrt{\epsilon}}{2}, \frac{\epsilon}{2(a+b)})$. Take $n_a \in \mathbb{Z}^+$ such that $n \geq n_a$ implies $|\sup(\{a_n, a_{n+1}, \dots\}) - a| < \epsilon'/2$ and $n_b \in \mathbb{Z}^+$ such that $n \geq n_b$ implies $|b_n - b| < \epsilon'$. Take some $n \in \mathbb{Z}^+$ such that $n \geq \max(n_a, n_b)$.

Denoting $s := \sup\{a_n, a_{n+1}, \dots\}$, we note by the definition of the supremum that we can find some $m \in \mathbb{Z}^+$ with $m \geq n$ and $|a_m - s| < \epsilon'/2$. By the triangle inequality,

$$|a_m - a| \leq |a_m - s| + |s - a| < \frac{\epsilon'}{2} + \frac{\epsilon'}{2} = \epsilon'$$

Accounting for the minimal case, we have

$$\begin{aligned}
\sup\{a_n b_n, a_{n+1} b_{n+1}, \dots\} &> a_m(b - \epsilon') \geq (a - \epsilon')(b - \epsilon') \\
&= ab - (a + b)\epsilon' + \epsilon'^2 \geq ab - \frac{\epsilon}{2} + \epsilon'^2 \\
&> ab - \frac{\epsilon}{2} > ab - \epsilon
\end{aligned}$$

On the other hand, we can account for the maximal case as

$$\begin{aligned}
\sup\{a_n b_n, a_{n+1} b_{n+1}, \dots\} &< s(b + \epsilon') < (a + \epsilon')(b + \epsilon') \\
&= ab + (a + b)\epsilon' + \epsilon'^2 \leq ab + \frac{\epsilon}{2} + \epsilon'^2 \\
&\leq ab + \frac{3\epsilon}{4} < ab + \epsilon
\end{aligned}$$

Hence, $|\sup\{a_n b_n, a_{n+1} b_{n+1}, \dots\} - ab| < \epsilon$. Therefore, $\limsup(a_n b_n) = ab$. ■

3. Show that $\lim n^{1/n} = 1$.

Proof. By the definition of the exponent, we have $n^{1/n} = e^{(\log n)/n}$ which is valid for $n \geq 1$. By l'Hôpital's rule, we have

$$\lim_{n \to \infty} \frac{\log n}{n} = \lim_{n \to \infty} \frac{1/n}{1} = 0$$

Hence, $\lim_{n \to \infty} e^{(\log n)/n} = e^0 = 1$. ■

§3: Analytic functions as mappings. Möbius transformations.

8. Let T be a Möbius transformation with the action $Tz = \frac{az+b}{cz+d}$. Show that $T(\mathbb{R}_\infty) = \mathbb{R}_\infty$ if and only if we can choose a, b, c, d to be real numbers.

Proof. First suppose $a, b, c, d \in \mathbb{R}$. Take some $x \in \mathbb{R}_\infty$ such that we have $Tx = \frac{ax+b}{cx+d} = y \in T(\mathbb{R}_\infty)$. Both the numerator and denominator are in $\mathbb{R}$ and hence $y \in \mathbb{R}_\infty$.

Now take some $y \in \mathbb{R}_\infty$. We want to find some $x \in \mathbb{R}_\infty$ such that $Tx = y$. We have two cases. If $a = cy$, then take $x = \infty$, giving $Tx = \frac{a}{c} = y$. Otherwise, let $x = \frac{dy-b}{a-cy}$. Then

$$Tx = \frac{a\frac{dy-b}{a-cy} + b}{c\frac{dy-b}{a-cy} + d} = \frac{ady - ab + ab - bcy}{cdy - bc + ad - cdy} = \frac{(ad-bc)y}{ad-bc} = y$$

since $ad - bc \neq 0$ for a Möbius transformation.

Conversely, suppose $T(\mathbb{R}_\infty) = \mathbb{R}_\infty$. Then $T0 = \frac{b}{d} \in \mathbb{R}_\infty$ and $T\infty = \frac{a}{c} \in \mathbb{R}_\infty$. We first assume that $c, d \neq 0$ so that we can write $a = cl$ and $b = kd$ for some $l, k \in \mathbb{R}$. Hence, for $x \in \mathbb{R}_\infty$ we have

$$Tx = \frac{clx + kd}{cx + d} = \frac{ckx + kd}{cx + d} + \frac{c(l-k)x}{cx+d} = k + m\frac{cx}{cx+d}$$

with $m := l - k \in \mathbb{R}$. We must have $m \neq 0$ for otherwise $l = \frac{b}{d} = \frac{a}{c}$ implies that $ad - bc = 0$, contradicting the definition of the Möbius transformation. We thus have $\frac{cx}{cx+d} \in \mathbb{R}_\infty$ for all $x \in \mathbb{R}_\infty$. Take some $x \in \mathbb{R}_\infty$ such that $x \neq 0, -d/c$. Decomposing $c := c_1 + c_2 i$ and $d := d_1 + d_2 i$, we thus have

$$\frac{cx}{cx+d} = \frac{c_1 x + c_2 xi}{c_1 x + d_1 + (c_2 x + d_2)i} \cdot \frac{(c_2 x - d_2)i}{(c_2 x - d_2)i} \tag{10}$$

$$= \frac{(c_1^2 + c_2^2)x^2 + (c_1 d_1 + c_2 d_2)x + (c_2 d_1 - c_1 d_2)xi}{|cx+d|^2} \tag{11}$$

so that we must have $c_2 d_1 = c_1 d_2$, thereby implying $d = pc$ for some $p \in \mathbb{R}$ by our restriction $c, d \neq 0$. Note that the denominator is non-zero by hypothesis. Therefore, $a = lc$, $b = kd = kpc$, $d = pc$, and $c = c$, so that all numbers are real multiples of c and hence each other. By Exercise 20, we can freely multiply each by $\bar{c}$, so that all are real.

We now return to the non-robust cases where $c = 0$ or $d = 0$. We cannot have both happen at the same time for otherwise $ad - bc = 0$. If $c = 0$, then $Tx = \frac{ax+b}{cx+d} = \frac{ax+b}{d}$. Evaluating $T0 = \frac{b}{d} \in \mathbb{R}$ gives $b = kd$ such that $Tx = \frac{ax+kd}{d} = k + \frac{ax}{d}$, and $T1 = k + \frac{a}{d}$ gives $a = md$ and we have a, b, c, d as real multiples of each other (with $c = 0$). If $d = 0$, then $Tx = \frac{ax+b}{cx} = \frac{a}{c} + \frac{b}{cx}$. Evaluating $T\infty = \frac{a}{c}$ gives $a = cl$ and hence $Tx = l + \frac{b}{cx}$. Evaluating $T1 = l + \frac{b}{c}$ gives $b = nc$ and hence we have a, b, c, d as real multiples of each other (with $d = 0$) once again and we are done. ∎

§IV Complex Integration

§1: Riemann-Stieltjes Integrals

1. Let $\gamma[a,b] \to \mathbb{R}$ be non-decreasing. Show that γ is of bounded variation and $V(\gamma) = \gamma(b) - \gamma(a)$.

Proof. Take some partition $P = \{a = t_0 < t_1 < \cdots < t_m = b\}$ of $[a,b]$. We have

$$v(\gamma; P) = \sum_{k=1}^{m} |\gamma(t_k) - \gamma(t_{k-1})| = \sum_{k=1}^{m} \gamma(t_k) - \gamma(t_{k-1}) = \gamma(b) - \gamma(a)$$

so that each path achieves the total variation $V(\gamma) = \gamma(b) - \gamma(a)$. ■

2. Let $\gamma[a,b] \to \mathbb{C}$ be of bounded variation. Prove that:

(a) If P and Q are partitions of $[a,b]$ and $P \subseteq Q$ then $v(\gamma; P) \leq v(\gamma, Q)$.

Proof. Iterate over Q until we find some t_n such that $t_n \notin P$. Relabel t_{n-1} as s_0 and the next element of P (it could be b) as s_l, where $l \geq 2$ by construction. The summation up to the index s_0 is identical for both P and Q. We will now show that the summation step from s_0 to s_l for Q must be greater than or equal to the equivalent step for P, which by definition occurs in a singular step $|\gamma(s_l) - \gamma(s_0)|$. To see this, we have by the triangle inequality

$$\begin{aligned}\sum_{k=1}^{l} |\gamma(s_k) - \gamma(s_{k-1})| &\leq |\gamma(s_1) - \gamma(s_0) + \gamma(s_2) - \gamma(s_1) + \cdots + \gamma(s_l) - \gamma(s_{l-1}| \\ &= |\gamma(s_l) - \gamma(s_{l-1})|.\end{aligned}$$

Thus, for each step in P we add an equal or greater sum for Q so that $v(\gamma; P) \leq v(\gamma, Q)$. ■

(b) If $\sigma : [a,b] \to \mathbb{C}$ is also of bounded variation and $\alpha, \beta \in \mathbb{C}$ then $\alpha\gamma + \beta\sigma$ is of bounded variation and $V(\alpha\gamma + \beta\sigma) \leq |\alpha|V(\gamma) + |\beta|V(\sigma)$.

Proof. We will show for any partition P that $v(\alpha\gamma + \beta\sigma; P) \leq |\alpha|V(\gamma) + |\beta|V(\sigma)$. Take some partition $P = \{a = t_0 < t_1 < \cdots < t_m = b\}$ of $[a,b]$. By the definition of the total variation,

$$\sum_{k=1}^{m} |\gamma(t_k) - \gamma(t_{k-1})| \leq V(\gamma),$$
$$\sum_{k=1}^{m} |\sigma(t_k) - \sigma(t_{k-1})| \leq V(\sigma).$$

Hence, by the triangle inequality and definition of absolute value,

$$\begin{aligned}
v(\alpha\gamma + \beta\sigma) &= \sum_{k=1}^{m} |\alpha\gamma(t_k) + \beta\sigma(t_k) - \alpha\gamma(t_{k-1}) - \beta\sigma(t_{k-1})| \\
&\leq \sum_{k=1}^{n} |\alpha\gamma(t_k) - \alpha\gamma(t_{k-1})| + |\beta\sigma(t_k) - \beta\sigma(t_{k-1})| \\
&= \sum_{k=1}^{n} |\alpha(\gamma(t_k) - \gamma(t_{k-1}))| + |\beta(\sigma(t_k) - \sigma(t_{k-1}))| \\
&\leq \sum_{k=1}^{n} |\alpha||\gamma(t_k) - \gamma(t_{k-1})| + |\beta||\sigma(t_k) - \sigma(t_{k-1})| \\
&= |\alpha| \sum_{k=1}^{n} |(\gamma(t_k) - \gamma(t_{k-1})| + |\beta| \sum_{k=1}^{n} |\sigma(t_k) - \sigma(t_{k-1})| \\
&\leq |\alpha| V(\gamma) + |\beta| V(\sigma).
\end{aligned}$$

We have shown that $\alpha\gamma + \beta\sigma$ is of bounded variation and by definition of the supremum we have $V(\alpha\gamma + \beta\sigma) \leq |\alpha|V(\gamma) + |\beta|V(\sigma)$ since we have already provided the latter upper bound. ■

7. Show that $\gamma : [0, 1] \to \mathbb{C}$, defined by $\gamma(t) = t + it \sin \frac{1}{t}$ for $t \neq 0$ and $\gamma(0) = 0$, is a path but is not rectifiable.

Proof. $\gamma(t)$ is continuous on $[0, 1]$ and hence a path if and only if its real and imaginary components are continuous on the same interval. Clearly the real component is continuous; for the imaginary component, it suffices to show that $\sin \frac{1}{x}$ is continuous on $(0, 1]$. This can be seen readily by taking the series expansion

$$\sin \frac{1}{x} = x^{-1} - \frac{x^{-3}}{3!} + \frac{x^{-5}}{5!} - \frac{x^{-7}}{7!} + \dots$$

which is clearly differentiable and hence continuous on $(0,1]$. Hence the function $it\sin\frac{1}{t}$ is continuous on $(0,1]$ and finally we have $\lim_{t\to 0^+} it\sin\frac{1}{t} = 0$ since for any $\epsilon > 0$ taking $\delta = \epsilon$ guarantees that for $0 < t < \delta$ we have $|t\sin\frac{1}{t}| \leq |t| < \epsilon$.

To see that γ is not rectifiable, observe that there are an infinite number of solutions to the equation $\sin\frac{1}{x} = 1$ of the form $x = \frac{2}{\pi}\frac{1}{1+4l}, l > 0 \in \mathbb{Z}$ with $0 < x < 1$ for all solutions, as well as infinite solutions to $\sin\frac{1}{x} = -1$ of the form $x = \frac{2}{\pi}\frac{1}{3+4l}$ with again $0 < x < 1$; it is easy to see that these solutions alternate and hence we can form a partition P by iterating alternatively, starting with $l = 0$ an arbitrary number of times, each sum contributing a total length 2. It is readily observed then that we can surpass any constant M in the summation and γ is not of bounded variation and hence not rectifiable. ■

22. Show that if F_1 and F_2 are primitives for $f : G \to \mathbb{C}$ and G is connected then there is a constant c such that $F_1(z) = c + F_2(z)$ for each z in G.

Proof. By definition, $F_1'(z) = f(z) = F_2'(z)$. Suppose there exists $\alpha, \beta \in G$ such that $F_1(\alpha) - F_2(\alpha) = c_\alpha$ and $F_1(\beta) - F_2(\beta) = c_\beta$. Then if we can find a rectifiable path γ whose trace $\{\gamma\}$ goes from α to β, we have

$$F_1(\beta) - F_1(\alpha) = \int_\gamma f = F_2(\beta) - F_2(\alpha),$$

such that $F_2(\beta) - F_1(\beta) = c_\beta = c_\alpha = F_2(\alpha) - F_1(\alpha)$ as desired. But since G is connected, there exists a polygon from α to β lying entirely in G, which we can use to define $\{\gamma\}$, and which is clearly of bounded variation and hence rectifiable. ■

§2: Power series representations of analytic functions

1. Let $\varphi : [a,b] \times [c,d] \to \mathbb{C}$ be a continuous function and define $g : [c,d] \to \mathbb{C}$ by

$$g(t) = \int_a^b \varphi(s,t)\,\mathrm{d}s.$$

Show that g is continuous.

Proof. Take some $t_0 \in [c,d]$. Fix $\epsilon < 0$. By the uniform continuity (uniform since $[a,b] \times [c,d]$ constitutes a compact set) of $\varphi(s,t)$, there exists some $\delta > 0$ such that $|t - t_0| < \delta$ implies that $|\varphi(s,t_0) - \varphi(s,t)| < \epsilon/(b-a)$ for any $s \in [a,b]$. Hence, for $|t - t_0| < \delta$, we have

$$
\begin{aligned}
|g(t) - g(t_0)| &= \left| \int_a^b \varphi(s,t)\,\mathrm{d}s - \int_a^b \varphi(s,t_0)\,\mathrm{d}s \right| = \left| \int_a^b [\varphi(s,t) - \varphi(s,t_0)]\,\mathrm{d}s \right| \\
&\leq \int_a^b |[\varphi(s,t) - \varphi(s,t_0)]\,\mathrm{d}s| = \int_a^b |\varphi(s,t) - \varphi(s,t_0)|\,\mathrm{d}s \\
&< (b-a)\frac{\epsilon}{b-a} = \epsilon
\end{aligned}
$$

so that $\lim_{t\to t_0} g(t) = g(t_0)$ and $g(t)$ is continuous. ■

2. Prove the following analogue of Leibniz's rule. Let G be an open set and let γ be a rectifiable curve in $\mathbb{C}$. Suppose that $\varphi : \{\gamma\} \times G \to \mathbb{C}$ is a continuous function and define $g : G \to \mathbb{C}$ by

$$
g(z) = \int_\gamma \varphi(w,z)\,\mathrm{d}w.
$$

Then g is continuous. If $\frac{\partial \varphi}{\partial z}$ exists for each (w,z) in $\{\gamma\} \times G$ and is continuous then g is analytic and

$$
g'(z) = \int_\gamma \frac{\partial \varphi}{\partial z}(w,z)\,\mathrm{d}w.
$$

Proof. Fix $\epsilon > 0$, take some path γ from the specified curve, and consider a point $z_0 \in G$. Since G is open, it has an open ball $B(z_0;\alpha) \subseteq G$ for some $\alpha > 0 \in \mathbb{R}$; the closed ball $H := \overline{B}(z_0,\beta)$ with $\beta := \alpha/2$ thus also has $H \subseteq G$. The set $\{\gamma\} \times H$ is compact, as can be seen by the fact that each of its members is a limit point. Thus, $\varphi : \{\gamma\} \times H \to \mathbb{C}$ is uniformly continuous. Therefore, there exists a δ of the form $0 < \delta < \beta$ such that if $|z - z_0| < \delta$, then $|\varphi(w,z) - \varphi(w,z_0)| < \epsilon/V(\gamma)$ for all $w \in \{\gamma\}$, where $V(\gamma)$ is characteristic of the curve and finite by γ as rectifiable. Hence, if $|z - z_0| < \delta$, we have

$$
\begin{aligned}
|g(z) - g(z_0)| &= \left| \int_\gamma \varphi(w,z)\,\mathrm{d}w - \int_\gamma \varphi(w,z_0)\,\mathrm{d}w \right| = \left| \int_\gamma [\varphi(w,z) - \varphi(w,z_0)]\,\mathrm{d}w \right| \\
&\leq \int_\gamma |\varphi(w,z) - \varphi(w,z_0)|\,|\mathrm{d}w| \leq V(\gamma)\,|\varphi(w,z) - \varphi(w,z_0)| \\
&< V(\gamma)\frac{\epsilon}{V(\gamma)} < \epsilon
\end{aligned}
$$

so that $\lim_{z\to z_0} g(z) = g(z_0)$. Since z_0 was chosen arbitrarily, we have that g is continuous on G.

We now consider the derivative of g. As in the original proof, it suffices simply to show that the derivative exists and is equal to the given form. Fix $\epsilon > 0$. Take some point $z_0 \in G$ and

construct a closed ball $H := \overline{B}(z_0, \beta$ for some β once again such that $z_0 \in H$. Then $\varphi_2 := \frac{\partial \varphi}{\partial z}$ is uniformly continuous on H. Hence there exists some $\delta > 0$ with $\delta < \beta$ such that if $|z - z_0| < \delta$, then $|\varphi_2(w, z) - \varphi_2(w, z_0)| < \epsilon / V(\gamma)$ for all $w \in \{\gamma\}$. Then if we form a line integral with the path σ given by the straight line from z_0 to z (valid by the existence of the closed ball) for some $w \in \{\gamma\}$, we have

$$\left| \int_\sigma [\varphi_2(w, z') - \varphi_2(w, z_0)] \, \mathrm{d}z' \right| \leq \int_\sigma |\varphi_2(w, z') - \varphi_2(w, z_0)| \, |\mathrm{d}z'| < \frac{\epsilon |z - z_0|}{V(\gamma)}.$$

Since σ is clearly rectifiable, we apply the Fundamental Theorem of Calculus for line integrals with the primitive $\Phi(z) = \varphi(w, z) - z\varphi_2(w, z_0)$ to conclude

$$\begin{aligned} \left| \int_\sigma [\varphi_2(w, z') - \varphi_2(w, z_0)] \, \mathrm{d}z' \right| &= \left| \varphi(w, z') - z' \varphi_2(w, z_0) \Big|_{z_0}^{z} \right| \\ &= |\varphi(w, z) - \varphi(w, z_0) - (z - z_0)\varphi_2(w, z_0)| \\ &< \frac{\epsilon |z - z_0|}{V(\gamma)} \end{aligned}$$

for any fixed $w \in \{\gamma\}$. Now integrating over γ,

$$\begin{aligned} \left| \int_\gamma \left[\frac{\varphi(w, z) - \varphi(w, z_0)}{z - z_0} - \varphi_2(w, z_0) \right] \mathrm{d}w \right| &= \left| \int_\gamma \frac{\varphi(w, z)}{z - z_0} \, \mathrm{d}w - \int_\gamma \frac{\varphi(w, z_0)}{z - z_0} \, \mathrm{d}w - \int_\gamma \varphi_2(w, z_0) \, \mathrm{d}w \right| \\ &= \left| \frac{g(z)}{z - z_0} - \frac{g(z_0)}{z - z_0} - \int_\gamma \varphi_2(w, z_0) \, \mathrm{d}w \right| \\ &= \left| \frac{g(z) - g(z_0)}{z - z_0} - \int_\gamma \varphi_2(w, z_0) \, \mathrm{d}w \right| \\ &< \frac{\epsilon}{V(\gamma)} V(\gamma) = \epsilon. \end{aligned}$$

This shows that $g'(z)$ exists and is given by

$$g'(z) = \int_\gamma \varphi_2(w, z_0) \, \mathrm{d}w,$$

so that g is analytic. Continuity of $g'(z)$ follows immediately from the continuity of φ_2 over the domain. ■

3. Suppose that γ is a rectifiable curve in $\mathbb{C}$ and φ is defined and continuous on $\{\gamma\}$. Use Exercise 2 to show that

$$g(z) = \int_\gamma \frac{\varphi(w)}{w-z}\,\mathrm{d}w$$

is analytic on $\mathbb{C} \setminus \{\gamma\}$ and

$$g^{(n)}(z) = n! \int_\gamma \frac{\varphi(w)}{(w-z)^{n+1}}\,\mathrm{d}w.$$

Proof. Consider the function $\psi := \{\gamma\} \times \mathbb{C} \setminus \{\gamma\} \to \mathbb{C}$ defined by $\psi(w,z) = \varphi(w)/(w-z)$. Then since $w - z \neq 0$ by our choice of domain, we have $\psi(w,z)$ as well defined and moreover continuous; since $\{\gamma\}$ is a compact and hence closed set, it follows that $\overline{\{\gamma\}} = \mathbb{C} \setminus \{\gamma\}$ must be open. Thus, with $G = \mathbb{C} \setminus \{\gamma\}$, and ψ acting as φ, we can apply Exercise 2 to conclude that $g'(z)$ is analytic on $\mathbb{C} \setminus \{\gamma\}$ and the derivative is given explicitly as

$$g^{(n)}(z) = \int_\gamma \psi(w,z)\,\mathrm{d}w = \int_\gamma \frac{\partial^n \psi(w,z)}{\partial z^n} = n! \int_\gamma \frac{\varphi(w)}{(w-z)^{n+1}}\,\mathrm{d}w$$

by a simple induction argument. ∎

5. Give the power series expansion of $\log z$ about $z = i$ and find its radius of convergence.

Proof. We assume the principal branch such that $\theta \in (-\pi, \pi)$ and use the fact that $\frac{\mathrm{d}}{\mathrm{d}z} \log z = z^{-1}$ to compute

$$\begin{aligned}
\log(z-i) &= \log i + \frac{1}{1!}\frac{1}{i}(z-i) + \frac{1}{2!}\frac{-1}{i^2}(z-i)^2 + \frac{1}{3!}\frac{1}{i^3}(z-i)^3 + \frac{1}{4!}\frac{-1}{i^4}(z-i)^4 + \frac{1}{5!}\frac{1}{i^5}(z-i)^5 + \dots \\
&= \frac{i\pi}{2} - i(z-i) + \frac{1}{2}(z-i)^2 + \frac{i}{3}(z-i)^3 - \frac{1}{4}(z-i)^4 - \frac{i}{5}(z-i)^5 \\
&= \frac{i\pi}{2} + \sum_{n=1}^{\infty} \frac{-i^n}{n}(z-i)^n.
\end{aligned}$$

By III Proposition 1.4, since we have

$$\lim_{n\to\infty} |a_n/a_{n+1}| = \lim_{n\to\infty} \frac{n+1}{n} = 1,$$

we have $R = 1$. This is in agreement with the fact that the logarithm is not defined at $z = 0$. ∎

7. Use the results of this section to evaluate the following integrals:

(a) $\int_\gamma \frac{e^{iz}}{z^2}\,\mathrm{d}z, \qquad \gamma(t) = e^{it}, \qquad 0 \le t \le 2\pi;$

Proof. We cannot proceed directly by Proposition 2.15 since $f := \frac{e^{iz}}{z^2}$ is not defined and hence not analytic at the origin, which is interior to the path. However, we can construct a finite number l of open balls at various points a along the contour of some radius such that 0 is not included and whose union covers $\{\gamma\}$. If we can show that f is analytic on $\mathbb{C} \setminus \{0\}$, then by Theorem 2.8 if follows that for a fixed open ball B_i, f it admits a power series $f_i(z) = \sum_{n=0}^{\infty} a_n(z-a)^n$ with $a_n = \frac{1}{n!}f^{(n)}(a)$ well defined. Such a power series admits a primitive of the form $F_i(z) = c + \sum_{n=1}^{\infty} n a_n (z-a)^n$ where $c \in \mathbb{C}$ is a constant. Then we have

$$\begin{aligned}\int_\gamma f\,\mathrm{d}z &= \sum_{i=1}^{l} \int_{\gamma_i} f\,\mathrm{d}z = \sum_{i=1}^{l} \int_{\gamma_i} f_i\,\mathrm{d}z \\ &= F_1(a_1) - F_1(a_0) + F_2(a_2) - F_2(a_1) + \cdots + F_l(a_l) - F_l(a_{l-1}) \\ &= F_l(a_l) - F_1(a_l) = F_l(0) - F_1(0) = 0,\end{aligned}$$

where $F_i(a_i) = F_{i+1}(a_i)$ since the points of overlap have two power series expansions f_i and f_{i+1} which must agree at each overlapping point, such that by the Fundamental Theorem of Calculus $F_i(a)$ and $F_{i+1}(a)$ can only differ up to a constant for all such overlapping a; we implicitly choose that constant to be 0.

To show f is analytic on $\mathbb{C} \setminus \{0\}$, it suffices to observe that e^{iz} and z^{-2} are both analytic on the same domain. It then follows that the integral is 0. ■

(b) $\int_\gamma \frac{\mathrm{d}z}{z-a}\,\mathrm{d}z, \qquad \gamma(t) = a + e^{it}, \qquad 0 \le t \le 2\pi;$

Proof. We proceed with the strategy as in (a) to define a finite number of overlapping open balls whose union contains γ and not a. Clearly, $f(z) = \frac{1}{z-a}$ is analytic on $\mathbb{C} \setminus \{a\}$ and hence it follows that the integral is 0 once again. ■

(c) $\int_\gamma \frac{\sin z}{z^3}\,\mathrm{d}z, \qquad \gamma(t) = e^{it}, \qquad 0 \le t \le 2\pi;$

Proof. The situation is analogous to the previous two; $\sin(z)/z^3$ is analytic on $\mathbb{C} \setminus \{0\}$ and so the integral is 0. ■

(d) $\int_\gamma \frac{\log z}{z^n}\,\mathrm{d}z, \qquad \gamma(t) = 1 + \frac{1}{2}e^{it}, \qquad 0 \le t \le 2\pi \text{ and } n \ge 0.$

Proof. Once again, $\log z/z^n$ is analytic, this time on $B(1;1/2)$, and we apply Proposition 2.15 to conclude that the integral is 0. ■

§3: Zeros of an analytic function

1. Let f be an entire function and suppose there is a constant M, and $R > 0$, and an integer $n \geq 1$ such that $|f(z)| \leq M|z|^n$ for $|z| > R$. Show that f is a polynomial of degree $\leq n$.

Proof. By f as an entire function, we can expand $f(z)$ in a power series as

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n$$

for arbitrary a and z in $\mathbb{C}$, where $a_n = \frac{1}{n!} f^{(n)}(a)$. In particular, take some $R' \in \mathbb{R}$, with $R' > 0$. Consider the expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - R - R')^n.$$

By f as an entire function, $f(z)$ is analytic on $B(R - R'; R')$. Moreover, by hypothesis $|f(z)| \leq M(R+2R')^n$ for all $z \in B(R-R'; R')$. Now fix $m > n$ and take some $a \in B(R-R'; R')$. By Cauchy's Estimate,

$$|f^{(m)}(a)| \leq \frac{(m+1)!M(R+2R')^n}{(R')^{m+1}}.$$

It is readily observed that for a fixed a, say $a = \frac{3R}{2}$, we can take $|f^{(m)}(a)|$ to be arbitrarily small as $R' \to \infty$, so that we must have $f^{(m)}(a) = 0$. We can easily find a dense collection of such a values which are zeros of the function $f^{(m)}$. It follows from Corollary 3.8 that $f^{(m)}$ must be 0. Since $m > n$ was arbitrary, it follows that f is a polynomial of degree $\leq n$, as was desired. ■

2. Given an example to show that G must be assumed to be connected in Theorem 3.7 (stated below).

Theorem. *Let G be an open connected set and let $f : G \to \mathbb{C}$ be an analytic function. Then the following are equivalent statements:*

(a) $f \equiv 0$;

(b) there is a point a in G such that $f^{(n)}(a) = 0$ for each $n \geq 0$;

(c) $\{z \in G : f(z) = 0\}$ has a limit point in G.

Proof. Take $G = B(0;1) \cup B(3;1)$. It is evident that G is open but not connected. Consider the function

$$f(z) = \begin{cases} 0 & \text{if } |z| < 1 \\ 1 & \text{if } |z| \geq 1. \end{cases}$$

We have $f^{(n)}(0) = 0$ for all $n \geq 0$, but $f \neq 0$, contradicting the theorem. ■

§5: Cauchy's Theorem

1. Suppose $f : G \to \mathbb{C}$ is analytic (G is an open subset of the plane) and define $\varphi : G \times G \to \mathbb{C}$ by $\varphi(z,w) = [f(z) - f(w)](z-w)^{-1}$ if $z \neq w$ and $\varphi(z,z) = f'(z)$. Prove that φ is continuous and for each fixed w, $z \to \varphi(w,z)$ is analytic.

Proof. Consider some $(z_0, w_0) \in G \times G$, where $z_0 \neq w_0$. We can find some $\delta > 0$ such that $B((z_0,w_0),\delta) \subseteq G$ has the property that for all $(w,z) \in B((z_0,w_0),\delta)$, we have $w \neq z$. Note that the distance function d is given as $d((z_0,w_0),(z_1,w_1)) = \sqrt{|z_0 - z_1|^2 + |w_0 - w_1|^2}$. It is a fact that the quotient of continuous functions is also continuous, in so far as the denominator is not zero, regardless of the metric space we are working with.

In particular, we use that f is analytic and hence continuous on G to conclude that clearly $f(z) - f(w)$ will be continuous on $G \times G$. By hypothesis the difference $z - w \neq 0$ and $z - w$ is also continuous by elementary means. Hence we have shown continuity except for points of the form (z,z).

Consider such a point (z_0, z_0). Fix $\epsilon > 0$. Since f is analytic and hence contains a continuous derivative, there exists some $\delta_1 > 0$ such that for $|z - z_0| < \delta_1$ we have $|f'(z) - f'(z_0)| < \epsilon/2$.

Moreover, by the differentiability of f we can find a δ_2 such that $|z - w| < \delta_2$ implies that

$$\left| \frac{f(z) - f(w)}{z - w} - f'(z) \right| < \frac{\epsilon}{2}.$$

Hence consider a point of the form (z,w) with the property that $|z - z_0| < \delta$ and $|z - w| < \delta$, where $\delta = \min(\delta_1, \delta_2)$. This can be easily done by taking $d((z,w),(z_0,z_0)) < \delta/2$, since

$$|z - z_0| \leq d((z,w),(z_0,z_0)) < \delta$$

and

$$\begin{aligned}|z-w| &\leq |z-z_0| + |w-z_0| = \sqrt{|z-z_0|^2 + 2|z-z_0||w-z_0| + |w-z_0|^2} \\ &\leq \sqrt{|z-z_0|^2 + 2|z-z_0||w-z_0| + |w-z_0|^2} \\ &\leq \sqrt{2|z-z_0|^2 + 2|w-z_0|^2} \\ &\leq \sqrt{2} d((z,w),(z_0,z_0)) < \delta.\end{aligned}$$

Hence, if $d((z,w),(z_0,z_0)) < \delta/2$, then we have two cases. If $z = w$, then we have

$$|\varphi(z,w) - \varphi(z_0,z_0)| = |f'(z) - f'(z_0)| < \frac{\epsilon}{2} < \epsilon.$$

If $z \neq w$, then

$$\begin{aligned}|\varphi(z,w) - \varphi(z_0,z_0)| &= \left| \frac{f(z)-f(w)}{z-w} - f'(z_0) \right| \\ &\leq \left| \frac{f(z)-f(w)}{z-w} - f'(z) \right| + |f'(z) - f'(z_0)| \\ &\leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.\end{aligned}$$

We have hence shown that φ is continuous on G. Now fix some $w \in G$. Consider the function $g(z) := \varphi(w,z)$. Consider some $z_0 \neq w$. Then we can find an open ball $B(z_0,\delta) \subseteq G$ for some $\delta > 0$ such that for $z \in B(z_0,\delta)$ we have $z \neq w$. Hence, $g(z) = [f(z) - f(w)](z-w)^{-1}$ on $B(z_0,\delta)$. To show g is analytic at z_0, we note by the analycity of f on G that clearly f is analytic on $B(z_0,\delta)$ and hence $f(z) - f(w)$ is analytic. Moreover, since $z \neq w$, it follows that $z - w$ is also analytic on $B(z_0,\delta)$. By the quotient rule, it follows that g is analytic on $B(z_0,\delta)$ and hence at z_0. Explicitly, for $z \neq w$,

$$g'(z) = \frac{f'(z)(z-w) - (f(z) - f(w))}{(z-w)^2}.$$

This argument covers all cases where $z_0 \neq w$. We now consider the derivative at w. By definition,

$$g'(w) = \lim_{z \to w} \frac{g(z) - g(w)}{z - w}.$$

We claim that $g'(w) = f''(w)/2$. Since f is analytic on G, consider the power series expansion about w, i.e.

$$f(z) = \sum_{n=0}^{\infty} a_n (z-w)^n = a_0 + a_1(z-w) + a_2(z-w)^2 + \cdots,$$

where in particular $a_0 = f(w)$, $a_1 = f'(w)$, and $a_2 = f''(w)/2$ by definition. Fix $\epsilon > 0$. We are looking for a $\delta > 0$ such that for $|z - w| < \delta$ we have

$$\begin{aligned}
\left|\frac{g(z) - g(w)}{z - w} - a_2\right| &= \left|\frac{\frac{f(z)-f(w)}{z-w} - f'(w)}{z - w} - a_2\right| \\
&= \left|\frac{\frac{a_0+a_1(z-w)+a_2(z-w)^2+a_3(z-w)^3+\cdots-a_0}{z-w} - a_1}{z - w} - a_2\right| \\
&= \left|\frac{a_1 + a_2(z-w) + a_3(z-w)^2 + \cdots - a_1}{z - w} - a_2\right| \\
&= |a_2 + a_3(z-w) + \cdots - a_2| \\
&= |a_3(z-w) + a_4(z-w)^2 + \cdots| \\
&= \left|\sum_{n=0}^{\infty} a_{n+3}(z-w)^{n+1}\right| := |b| = \left|\sum_{n=1}^{\infty} b_n(z-w)^n\right| < \epsilon.
\end{aligned}$$

We note that since b_n has coefficients of absolute value strictly smaller than the expansion $f''(z)$ (which inherits the radius of convergence R of $f(z)$), it follows that the radius of convergence R' of b_n satisfies $R' \geq R$. Since $R' > 0$, we have $1/R' \in \mathbb{R}$. Note that for our purposes $R' = \infty$ does not pose any problems, taking $1/R' = 0$. Set $\epsilon' = \frac{2}{R'}$. By the definition of the radius of convergence R', there exists some $N \in \mathbb{Z}^+$ such that for $n \geq N$, we have

$$\frac{1}{2R'} < \sup\{|b_n|^{1/n}, |b_{n+1}|^{1/n+1}, \dots\} < \frac{3}{2R'} := \sigma_1.$$

That is, for $n \geq N$, $\sigma_1^n > |b_n|$. Now since $N \in \mathbb{Z}^+$, and hence there are a finite number of terms N which are not strictly within the specified tolerance, there exists a σ_2 such that for $n < N$, we have $\sigma_2^n > |b_n|$. Take $\sigma = \max(\sigma_1, \sigma_2)$. Then we have

$$\left|\sum_{n=1}^{\infty} b_n(z-w)^n\right| \leq \sum_{n=1}^{\infty} |b_n||z-w|^n \leq |b_1| \sum_{n=1}^{\infty} (\sigma|z-w|)^n = \frac{|b_1|}{1 - \sigma|z-w|}$$

by the geometric series formula, for which $\delta < \epsilon(|b_1| + \epsilon\sigma)^{-1}$ yields a summation of less than ϵ. Hence, $g'(w) = f''(w)/2$, as desired. We have shown that for arbitrary w, the function $z \to \varphi(w, z)$ is differentiable on G. Although we have not shown that the derivative is continuous, we note that this follows immediately from differentiability within the complex context, although it could also be shown directly with a similar ϵ-δ argument. ■

4. Show that the Integral Formula follows from Cauchy's Theorem.

Proof. Take some open $G \subseteq \mathbb{C}$ and consider some $f : G \to \mathbb{C}$ which is analytic. Take some curve γ such that $n(\gamma; w) = 0$ for all $w \in \mathbb{C} \setminus G$. Fix some $a \in G \setminus \{\gamma\}$. Now consider the function

$$g(z) := \frac{f(z) - f(a)}{z - a},$$

defined on $\{\gamma\}$. Since $a \notin \{\gamma\}$, it follows that $z - a \neq 0$. Since the numerator and denominator are both analytic, and $z - a \neq 0$, it follows from the quotient rule that $g(z)$ is analytic on $\{\gamma\}$. Then by Cauchy's Theorem,

$$\int_\gamma g \, \mathrm{d}z = \int_\gamma \frac{f(z) - f(a)}{z - a} \, \mathrm{d}z = 0.$$

Noting that $\int_\gamma f(a)/(z-a) \, \mathrm{d}z = f(a) \int_\gamma (z-a)^{-1} \, \mathrm{d}z$ must converge since there exists a minimal value of $|z - a|$ which is greater than 0, it follows that we may separate the integral as

$$\int_\gamma \frac{f(z) - f(a)}{z - a} \, \mathrm{d}z = \int_\gamma \frac{f(z)}{z - a} \, \mathrm{d}z - \int_\gamma \frac{f(a)}{z - a} \, \mathrm{d}z = 0$$

and hence

$$\int_\gamma \frac{f(z)}{z - a} \, \mathrm{d}z = \int_\gamma \frac{f(a)}{z - a} \, \mathrm{d}z = f(a) \int_\gamma \frac{1}{z - a} \, \mathrm{d}z = 2\pi i f(a) n(\gamma; a),$$

from which diving through by $2\pi i$ gives

$$\frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z - a} \, \mathrm{d}z = n(\gamma; a) f(a),$$

which is Cauchy's Integral Formula, as desired. ■

5. Let γ be a rectifiable curve in $\mathbb{C}$ and $a \notin \gamma$. Show that for $n \geq 2$, $\int_\gamma (z - a)^{-n} \, \mathrm{d}z = 0$.

Proof. This is a direct corollary of Theorem 5.8. Consider the function $f : \mathbb{C} \to \mathbb{C}$ given by $f(z) = 1$. Note that by definition $\mathbb{C}$ is an open subset of $\mathbb{C}$. Since $\mathbb{C} \setminus \mathbb{C} = \varnothing$, the condition that $n(\gamma; w) = 0$ for all $win\mathbb{C} \setminus G$, where G is the input space of f, is vacuously true. For $k \geq 1$, we have

$$f^{(k)}(a) n(\gamma; a) = 0 \cdot n(\gamma; a) = 0 = \frac{k!}{2\pi i} \int_\gamma (z - a)^{-k-1} \, \mathrm{d}z.$$

Since $k! \neq 0$, it follows that for $n \geq 2$ we have $\int_\gamma (z - a)^{-n} \, \mathrm{d}z = 0$, as desired. ■

§6: The homotopic version of Cauchy's Theorem and simple connectivity.

1. Let G be a region and let $\sigma_1, \sigma_2 : [0,1] \to G$ be the constant curves $\sigma_1(t) \equiv a$, $\sigma_2(t) \equiv b$. Show that if γ is a closed, rectifiable curve in G and $\gamma \sim \sigma_1$ then $\gamma \sim \sigma_2$.

Proof. We will show that $\sigma_1 \sim \sigma_2$ and conclude by transitivity that $\gamma \sim \sigma_2$. Since G is a region, there exists a polygonal path which connects a to b with n steps for some $n \in \mathbb{Z}$ with $n \geq 1$. Define the path $\gamma : I \to G$ as that polygonal path, with each step i over the interval $[\frac{i-1}{n}, \frac{i}{n}]$ with $i \in \{1, \ldots, n\}$ a linear path connecting the two points in consideration.

Now consider the map $\Gamma : I^2 \to G$ by $\Gamma(s,t) = \gamma(t)$. Observe that by γ as a polygonal path we have $\gamma(t) \in G$ for all $t \in [0,1]$ so that the function is well-defined. We have $\Gamma(s,0) = \gamma(0) = a$ and $\Gamma(s,1) = \gamma(1) = b$. Moreover, $\Gamma(0,t) = \gamma(t) = \Gamma(1,t)$ so that $\sigma_1 \sim \sigma_2$ and hence $\gamma \sim \sigma_2$. ■

2. Show that if we remove the requirement "$\Gamma(0,1) = \Gamma(1,t)$ for all t" from Definition 6.1 (shown below) then the curve $\gamma_0(t) = e^{2\pi i t}$, $0 \leq t \leq 1$, is homotopic to the constant curve $\gamma_1(t) \equiv 1$ in the region $G = \mathbb{C} - \{0\}$.

Definition. *Let $\gamma_0, \gamma_1 : I \to G$ be two closed rectifiable curves in a region G; then γ_0 is homotopic to γ_1 in G if there is a continuous function $\Gamma : I^2 \to G$ such that*

$$\begin{cases} \Gamma(s,0) = \gamma_0(s) \quad \textit{and} \quad \Gamma(s,1) = \gamma_1(s) \quad (0 \leq s \leq 1) \\ \Gamma(0,t) = \Gamma(1,t) \quad (0 \leq t \leq 1). \end{cases}$$

Proof. Consider the map $\Gamma : I^2 \to G$ given by $\Gamma(s,t) = e^{2\pi i s(1-t)} = e^{2\pi i s} e^{-2\pi i s t}$. This function is differentiable and hence continuous on G. We have $\Gamma(s,0) = e^{2\pi i s} = \gamma_0(s)$ and $\Gamma(s,1) = e^0 = 1 = \gamma_1(s)$ as desired. However, it is evident that $\Gamma(0,t) = e^0 = 1$ is equivalent to $\Gamma(1,t) = e^{2\pi i(1-t)}$ if and only if $t = k$ for $k \in \mathbb{Z}$, so that evidently they are not equivalent in general for $t \in [0,1]$. ■

3. Let $\mathscr{C}$ = all rectifiable curves in G joining a to b and show that Definition 6.11 (stated below) gives an equivalence relation on $\mathscr{C}$.

Definition. *If $\gamma_0, \gamma_1 : [0,1] \to G$ are two rectifiable curves in G such that $\gamma_0(0) = \gamma_1(0) = a$ and $\gamma_0(1) = \gamma_1(1) = b$ then γ_0 and γ_1 are fixed-end-point homotopic (FEP homotopic) if there is a continuous map $\Gamma : I^2 \to G$ such that*

$$\begin{aligned} \Gamma(s,0) &= \gamma_0(s) & \qquad \Gamma(s,1) &= \gamma_1(s) \\ \Gamma(0,t) &= a & \qquad \Gamma(1,t) &= b \end{aligned}$$

for $0 \leq s, t \leq 1$.

Proof. Consider arbitrary rectifiable curves $\gamma_0, \gamma_1, \gamma_2 : [0,1] \to G$ with $\gamma_0(0) = \gamma_1(0) = \gamma_2(0) = a$ and $\gamma_0(1) = \gamma_1(1) = \gamma_2(1) = b$. We will take $\sim$ to indicate a FEP homotopic relationship.

We first show reflexivity. Define $\Gamma : I^2 \to G$ by $\Gamma(s,t) = \gamma_0(s)$ for $0 \leq s, t \leq 1$. It follows readily from the continuity of $\gamma_0 : [0,1] \to \{\gamma\} \subseteq G$ that Γ is continuous on I^2 and hence $\gamma_0 \sim \gamma_0$.

Next we show symmetry. Suppose that $\gamma_0 \sim \gamma_1$. Then there exists some continuous map $\Gamma : I^2 \to G$ satisfying the FEP homotopic equations; consider the map $\Lambda : I^2 \to G$ given by $\Lambda(s,t) = \Gamma(s, 1-t)$. Clearly this map is continuous as well, for example by reparameterization. It follows that $\Lambda(s,0) = \gamma_1(s)$ and that $\Lambda(s,1) = \gamma_0(s)$, as desired. Moreover, $\Lambda(0,t) = \Gamma(0,1-t) = a$ and $\Lambda(1,t) = \Gamma(1,1-t) = b$ since $1 - t \in [0,1]$. Hence, $\gamma_1 \sim \gamma_0$.

Finally, we show transitivity. Suppose that $\gamma_0 \sim \gamma_1$ and $\gamma_1 \sim \gamma_2$ and define corresponding maps as Γ and Λ. We consider the map defined in the book as $\Phi : I^2 \to G$ with

$$\Phi(s,t) = \begin{cases} \Gamma(s,2t) & \text{if } 0 \leq t \leq 1/2 \\ \Lambda(s,2t-1) & \text{if } 1/2 \leq t \leq 1. \end{cases}$$

This map is well defined since $\Phi(s,1/2) = \Gamma(s,1/2) = \gamma_1(s) = \Lambda(s,0)$; this agreement coupled with the continuity of Γ and Λ on the domains indicated in Φ yields continuity of Φ. We have $\Phi(s,0) = \Gamma(s,0) = \gamma_0(s)$ and $\Phi(s,1) = \Lambda(s,1) = \gamma_2(s)$ as desired. Moreover, since $2t \in I$ for $t \in [0,1/2]$ and $2t - 1 \in I$ for $t \in [1/2,1]$, we have for $t \in [0,1/2]$ that $\Phi(0,t) = \Gamma(0,2t) = a$ and for $t \in [1/2,1]$ that $\Phi(0,t) = \Lambda(0,2t-1) = b$ which shows that $\gamma_0 \sim \gamma_2$. Thus, $\sim$ is an equivalence relation. ∎

5. Evaluate the integral $\int_\gamma \frac{\mathrm{d}z}{z^2+1}$ where $\gamma(\theta) = 2|\cos 2\theta| e^{i\theta}$ for $0 \leq \theta \leq 2\pi$.

Proof. Define $f : \mathbb{C} \setminus \{i, -i\} \to C$ by $f(z) = (z^2+1)^{-1}$, where we have removed i and $-i$ from the domain since $f(i) = f(-i) = \infty \notin \mathbb{C}$ if we would define $f(i)$ or $f(-i)$. Note then that taking $G = \mathbb{C} \setminus \{i\}$ yields that $\mathbb{C}_\infty \setminus G = \{i\} \cup \{-i\} \cup \{\infty\}$ which is not connected and hence G is not simply connected. In particular, we cannot immediately conclude that the integral vanishes via Cauchy's Theorem (Fourth Version).

Note that γ is not differentiable and hence not smooth at $\pi/4$, $3\pi/4$, $5\pi/4$, and $7\pi/4$. However, we can form the partition $[0, \pi/4, 3\pi/4, 5\pi/4, 7\pi/4, 2\pi]$ on θ and note that γ is piecewise smooth on each corresponding interval since we may safely collapse the absolute value to a positive or negative except at the boundary, which is irrelevant to the smoothness of the interval. Noting that the integrals in consideration are independent of 2π translation of θ, we may join the first and last intervals, which combine to form a piecewise smooth interval. We thus have the intervals $[\pi/4, 3\pi/4]$, $[3\pi/4, 5\pi/4]$, $[5\pi/4, 7\pi/4]$, and $[-\pi/4, \pi/4]$ with the corresponding paths γ_0, γ_1, γ_2, and γ_3. Hence,

$$\int_\gamma \frac{\mathrm{d}z}{z^2+1} = \int_{\gamma_0} \frac{\mathrm{d}z}{z^2+1} + \int_{\gamma_1} \frac{\mathrm{d}z}{z^2+1} + \int_{\gamma_2} \frac{\mathrm{d}z}{z^2+1} + \int_{\gamma_3} \frac{\mathrm{d}z}{z^2+1} = \int_{\gamma_0} \frac{\mathrm{d}z}{z^2+1} + \int_{\gamma_2} \frac{\mathrm{d}z}{z^2+1},$$

where we have evaluated the integrals about γ_1 and γ_3 as 0 by forming a disk with center a at -1, $-i$, and 1, respectively, with radius r satisfying $1 < r < \sqrt{2}$, and noting that since f is analytic on $B(a;r)$ it follows immediately (Proposition 2.15) that the integral over each closed, rectifiable path must vanish.

We will now show that $\int_{\gamma_0} \frac{\mathrm{d}z}{z^2+1} = -\int_{\gamma_2} \frac{\mathrm{d}z}{z^2+1}$. Noting that we may collapse the absolute value about γ to a negative in both cases, by the definition of the line integral we have

$$\int_{\gamma_0} \frac{\mathrm{d}z}{z^2+1} = \int_{\pi/4}^{3\pi/4} \frac{\mathrm{d}\theta}{(-2\cos 2\theta e^{i\theta})^2+1} \frac{\mathrm{d}\gamma(\theta)}{\mathrm{d}\theta} = \int_{\pi/4}^{3\pi/4} \mathrm{d}\theta \frac{2e^{i\theta}[2\sin 2\theta - i\cos 2\theta]}{4\cos^2 2\theta e^{2i\theta}+1}$$

and

$$\int_{\gamma_2} \frac{\mathrm{d}z}{z^2+1} = \int_{5\pi/4}^{7\pi/4} \frac{\mathrm{d}\theta}{(-2\cos 2\theta e^{i\theta})^2+1} \frac{\mathrm{d}\gamma(\theta)}{\mathrm{d}\theta} = \int_{5\pi/4}^{7\pi/4} \mathrm{d}\theta \frac{2e^{i\theta}[2\sin 2\theta - i\cos 2\theta]}{4\cos^2 2\theta e^{2i\theta}+1}.$$

Fix $\theta \in [\pi/4, \pi/2]$ and consider $\phi := \theta + \pi \in [5\pi/4, 7\pi/4]$. Consider the integrands of the integral over γ_0 and γ_2 at θ and ϕ, respectively; we have

$$\begin{aligned}
&\frac{2e^{i\theta}[2\sin 2\theta - i\cos 2\theta]}{4\cos^2 2\theta e^{2i\theta}+1} + \frac{2e^{i\phi}[2\sin 2\phi - i\cos 2\phi]}{4\cos^2 2\phi e^{2i\phi}+1} \\
=&\frac{2e^{i\theta}[2\sin 2\theta - i\cos 2\theta]}{4\cos^2 2\theta e^{2i\theta}+1} + \frac{2e^{i(\pi+\theta)}[2\sin 2(\pi+\theta) - i\cos 2(\pi+\theta)]}{4\cos^2 2(\pi+\theta) e^{2i(\pi+\theta)}+1} \\
=&\frac{2e^{i\theta}[2\sin 2\theta - i\cos 2\theta]}{4\cos^2 2\theta e^{2i\theta}+1} + \frac{-2e^{i\theta}[2\sin 2\theta - i\cos 2\theta]}{4\cos^2 2\theta e^{2i\theta}+1} \\
=&0.
\end{aligned}$$

Hence, we may pair up each element of the integral over γ_0 with a corresponding element in γ_1 in a bijective manner such that they sum to 0. Since the integrands are continuous as a combination of continuous constituents, each integral converges and we conclude that the sum of the integrals is 0. Hence, $\int_\gamma \frac{\mathrm{d}z}{z^2+1} = 0$. ■

§7: Counting zeros; the Open Mapping Theorem

1. Show that if $f : G \to \mathbb{C}$ is analytic in the region G and γ is a rectifiable curve in G, then $f \circ \gamma$ is also a rectifiable curve.

Proof. We begin with a lemma.

Lemma. *Suppose $f : G \to \mathbb{C}$ is analytic where G is open and $X \subseteq G$ is compact; then f is uniformly differentiable on X.*

Proof. We first note that the lemma could be somewhat more general by abstracting from $\mathbb{C}$ to an arbitrary metric space; we will assume that $X \subseteq \mathbb{C}$. It may also be the case that it is not necessary to require G as open. The lemma and consequent proof is strongly analogous to Section II Theorem 5.15.

Fix $\epsilon > 0$. We wish to find a $\delta > 0$ such that for any $w, z \in X$ which satisfy $|w - z| < \delta$, we have $\left|\frac{f(w)-f(z)}{w-z} - f'(z)\right| < \epsilon$. Suppose towards a contradiction that no such δ exists. Then for each $n \in \mathbb{Z}^+$, there exists $z_n, w_n \in X$ with $|w_n - z_n| < 1/n$ but $\left|\frac{f(w_n)-f(z_n)}{w_n-z_n} - f'(z_n)\right| \geq \epsilon$.

Since X is compact, there exists a subsequence $\{z_{n_k}\}$ and point $z \in X$ such that $z = \lim z_{n_k}$. We have

$$|z - w_{n_k}| \leq |z - z_{n_k}| + |z_{n_k} - w_{n_k}| \leq |z - z_{n_k}| + \frac{1}{n_k},$$

and since the right hand side tends to 0 at sufficiently high k we have that $z = \lim w_{n_k}$. Fix some $\epsilon' < \epsilon$. Hence,

$$\begin{aligned}
\epsilon &\leq \left|\frac{f(w_{n_k}) - f(z_{n_k})}{w_{n_k} - z_{n_k}} - f'(z_{n_k})\right| \\
&= \left|\frac{f(w_{n_k}) - f(z_{n_k})}{w_{n_k} - z_{n_k}} - f'(z) + f'(z) - f'(z_{n_k})\right| \\
&\leq \left|\frac{f(w_{n_k}) - f(z_{n_k})}{w_{n_k} - z_{n_k}} - f'(z)\right| + |f'(z) - f'(z_{n_k})| \\
&\leq \left|\frac{f(w_{n_k}) - f(z_{n_k})}{w_{n_k} - z_{n_k}} - f'(z)\right| + \frac{\epsilon'}{2},
\end{aligned}$$

where we have performed the last substitution by taking a sufficiently large $k := k_1$ by the continuity of f. Note that ϵ' may be considered independent of ϵ. Let $\lambda_{n_k} := w_{n_k} - z$ and $\tau_{n_k} := w_{n_k} - z$. Since f is analytic in G, it is permissible to perform a power series expansion at all $z' \in G$ with a corresponding radius of convergence R satisfying $0 < R \leq \infty$. Performing a power series expansion about z, and denoting the coefficients as a_j for $j \geq 0$,

$$
\begin{aligned}
&\left|\frac{f(w_n)-f(z_n)}{w_n-z_n}-f'(z)\right| \\
&=\left|\frac{\left(f(z)+f'(z)\lambda_{n_k}+\frac{f''(z)\lambda_{n_k}^2}{2!}+\cdots\right)-\left(f(z)+f'(z)\tau_{n_k}+\frac{f''(z)\tau_{n_k}^2}{2!}+\cdots\right)}{\lambda_{n_k}-\tau_{n_k}}-f'(z)\right| \\
&=\left|\frac{f'(z)(\lambda_{n_k}-\tau_{n_k})+\frac{f''(z)\left(\lambda_{n_k}^2-\tau_{n_k}^2\right)}{2!}+\cdots}{\lambda_{n_k}-\tau_{n_k}}-f'(z)\right| \\
&=\left|\frac{\frac{f''(z)\left(\lambda_{n_k}^2-\tau_{n_k}^2\right)}{2!}+\frac{f''(z)\left(\lambda_{n_k}^3-\tau_{n_k}^3\right)}{3!}+\cdots}{\lambda_{n_k}-\tau_{n_k}}\right|=\left|\frac{\sum_{j=2}^{\infty}\frac{\left(\lambda_{n_k}-\tau_{n_k}\right)}{j!}\sum_{l=0}^{j-1}\lambda_{n_k}^l\tau_{n_k}^{j-1-l}}{\lambda_{n_k}-\tau_{n_k}}\right| \\
&=\left|\sum_{j=2}^{\infty}a_j\sum_{l=0}^{j-1}\lambda_{n_k}^l\tau_{n_k}^{j-1-l}\right|\leq\sum_{j=2}^{\infty}\left|a_j\sum_{l=0}^{j-1}\lambda_{n_k}^l\tau_{n_k}^{j-1-l}\right|\leq\sum_{j=2}^{\infty}|a_j|\sum_{l=0}^{j-1}|\lambda_{n_k}|^l\,|\tau_{n_k}|^{j-1-l} \\
&\leq\sum_{j=2}^{\infty}|a_j|\,(|\lambda_{n_k}|+|\tau_{n_k}|)^{j-1}=\sum_{j=2}^{\infty}|a_j|\,\phi^{j-1},
\end{aligned}
$$

where $\phi := |\lambda_{n_k}|+|\tau_{n_k}|$ can be made arbitrarily small. By the existence of the radius of convergence R at z, it follows by a comparison to a geometric series analogous to §5, Exercise 1, that $\sum_{j=2}^{\infty}|a_j|\,\phi^{j-1}<\epsilon'/2$ for a sufficiently large $k := k_2$. Taking $k=\max(k_1,k_2)$, we hence have

$$
\epsilon\leq\left|\frac{f(w_{n_k})-f(z_{n_k})}{w_{n_k}-z_{n_k}}-f'(z)\right|+\frac{\epsilon'}{2}<\frac{\epsilon'}{2}+\frac{\epsilon'}{2}=\epsilon'<\epsilon,
$$

which is a contradiction. Hence, f is uniformly differentiable. ■

We now return to the original problem. First assume G is a disk $B(a;R)$. Since $\{\gamma\}$ is a compact subset of G, there exists some $d=\operatorname{dist}(\{\gamma\},\partial G)>0$ so that $\{\gamma\}\subseteq\overline{B}(a;\rho)\subseteq B(a;R)=G$ where $\rho=R-d/2$. Observe that $H:=\overline{B}(a;\rho)$ is clearly compact since every infinite set within must contain a limit point (II Theorem 4.9), we have by the lemma, that f is uniformly differentiable on H.

Fix $\epsilon>0$. By the uniform differentiability of f on H, there exists some $\delta_1>0$ such that for $w,z\in\{\gamma\}\subseteq H$ with $|w-z|<\delta_1$, we have

$$
\left|\frac{f(w)-f(z)}{w-z}-f'(z)\right|<\epsilon.
$$

Moreover, since γ is continuous and $[a,b]$ is compact, we have by II Theorem 5.15 that γ is uniformly continuous. Hence, there exists some $\delta_2 > 0$ such that $t, s \in [a,b]$ satisfying $|t-s| < \delta_2$ gives $|\gamma(t) - \gamma(s)| < \delta_1$. In particular, $|s-t| < \delta_2$ implies $|f(\gamma(t)) - f(\gamma(s))| < \epsilon$.

Consider some partition $P := \{a = t_0, t_1, \ldots, t_m = b\}$ of $[a,b]$ where $t_0 < t_1 < \cdots < t_m$ and $m \in \mathbb{Z}^+$. We may further subdivide this partition to P' in a finite number of steps such that for $t_i \in P'$ we have $t_i - t_{i-1} < \delta_2$. We can write $p' = \{a = s_0, s_1, \ldots, s_n = b\}$ for some $n \in \mathbb{Z}^+$. By the triangle inequality,

$$v(f \circ \gamma, P) = \sum_{k=1}^{m} |f(\gamma(t_k)) - f(\gamma(t_{k-1}))| \leq \sum_{k=1}^{n} |f(\gamma(s_k)) - f(\gamma(s_{k-1}))| = v(f \circ \gamma, P').$$

Since $f'(z)$ is continuous on H, by II Corollary 5.13 there exists some $z_0 \in H$ such that $|f'(z_0)| = \sup\{|f'(z)| : z \in H\}$. In particular, there exists some $M \in \mathbb{R}$ such that $f'(z) < M$ for all $z \in \{\gamma\}$. Hence, for a fixed s_k, we have

$$\begin{aligned}
&\left| \frac{f(\gamma(s_k)) - f(\gamma(s_{k-1}))}{\gamma(s_k) - \gamma(s_{k-1})} - f'(\gamma(t_k)) \right| < \epsilon \\
\implies &|f(\gamma(s_k)) - f(\gamma(s_{k-1})) - (\gamma(s_k) - \gamma(s_{k-1}))f'(\gamma(s_k))| < \epsilon |\gamma(s_k) - \gamma(s_{k-1})| \\
\implies &|f(\gamma(s_k)) - f(\gamma(s_{k-1}))| < |\gamma(s_k) - \gamma(s_{k-1})|\, f'(\gamma(s_k)) + \epsilon\, |\gamma(s_k) - \gamma(s_{k-1})| \\
\implies &|f(\gamma(s_k)) - f(\gamma(s_{k-1}))| < |\gamma(s_k) - \gamma(s_{k-1})|\, (f'(\gamma(s_k)) + \epsilon) < |\gamma(s_k) - \gamma(s_{k-1})|\, (M + \epsilon)\,.
\end{aligned}$$

Hence,

$$\begin{aligned}
v(f \circ \gamma, P) &\leq v(f \circ \gamma, P') \\
&= \sum_{k=1}^{n} |f(\gamma(s_k)) - f(\gamma(s_{k-1}))| \\
&< \sum_{k=1}^{n} |\gamma(s_k) - \gamma(s_{k-1})|\, (M + \epsilon) \\
&= (M + \epsilon) \sum_{k=1}^{n} |\gamma(s_k) - \gamma(s_{k-1})| \\
&= (M + \epsilon) v(\gamma, P') \\
&\leq (M + \epsilon) V(\gamma).
\end{aligned}$$

Since $V(\gamma) \in \mathbb{R}$, it follows that $(M + \epsilon)V(\gamma)$ may be taken arbitrarily close to $MV(\gamma)$. In particular, $v(f \circ \gamma, P)$ is bounded, and since P was arbitrary we have $V(f \circ \gamma) \leq MV(\gamma)$ and hence $f \circ \gamma$ is of bounded variation.

We next remove the requirement that G is a disk. Since $\{\gamma\}$ is compact, there exists some $r \in \mathbb{R}$ with $0 < r < \operatorname{dist}(\{\gamma\}, \partial G)$. By the absolute convergence of γ, there exists some $\delta > 0$ such that for $t, s \in [a, b]$ with $|t - s| < \delta$, we have $|\gamma(s) - \gamma(t)| < r/2$.

Take some partition P of $[a, b]$. We may without loss of generality assume that for $t_k, t_{k-1} \in P$ we have $t_k - t_{k-1} < \delta$ since passing to a smaller partition only increases the sum. Consider the open ball $B(\gamma(a); r \subseteq G$. Since f is analytic on G, it is also analytic on this disk. There exists some $m \in \mathbb{Z}^+$ such that $t_0, \ldots, t_m \in B(\gamma(a); r \subseteq G$. Consider the path σ defined as $\sigma(t) = \gamma(t)$ for $t \in [t_0, t_m]$. Since σ is clearly rectifiable on the disk, it follows from our considerations of disks that $f \circ \sigma$ is rectifiable with

$$\begin{aligned}\sum_{i=1}^{m} |f(\gamma(t_i)) - f(\gamma(t_{i-1}))| &= \sum_{i=1}^{m} |f(\sigma(t_i)) - f(\sigma(t_{i-1}))| \\ &= v(f \circ \sigma, P) \\ &\leq V(f \circ \sigma) \\ &\leq MV(\sigma) \\ &\leq MV(\gamma)\end{aligned}$$

where $M \in \mathbb{R}$ is defined as the maximal value of f' on $\{\gamma\}$, which is well-defined since $\{\gamma\}$ is compact. In particular, it must be the case that $|\gamma(t_m) - \gamma(t_0)| > r/2$ since otherwise by our choice of δ we would have $\gamma(t_{m+1}) \in B(\gamma(a); r$, contradicting the maximal nature of m. We can then repeat the process with t_m acting as t_0 and a maximal t'_m acting as t_m. By the rectifiable nature of γ, it follows that this process will terminate with a maximum of $\lceil V(\gamma)/r/2 \rceil = \lceil 2V(\gamma)/r \rceil$ separate intervals. It follows that the total sum $v(f \circ \gamma, P)$ satisfies $v(\gamma, P) \leq M\lceil 2V(\gamma)/r \rceil = R$ for some $R \in \mathbb{R}$ which is independent of the particular partition P, and hence that $f \circ \gamma$ is rectifiable. ■